



Transfusion Policy Manual

Warfarin Reversal: Management of an elevated INR in adult patients, with or without bleeding

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	All Clinical Staff

Introduction

In patients taking warfarin, an INR (International Normalised Ratio) greater than or equal to 5 significantly increases the risk of bleeding. There are several options available to reduce the INR. The appropriate option is dependent upon the urgency of INR reduction/normalisation, or the patient's risk of bleeding/bruising and the time to surgery/procedure, if any planned.

Risk Statement

Non-compliance with this guideline will

Breach legislative requirements	☐	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☐	Misconduct	☐
Breach SCGOPHCG Mission & Values	☒	Staff Safety	☐
Other:	☐		

Guideline Principles

This guideline provides general guidance for warfarin reversal. The dosage and management of therapy depends on patient factors, the location and extent of bleeding (if any), and the patient's clinical condition. For specialist advice contact the on-call haematologist/haematology registrar.

Human Prothrombin Complex Concentrates (PCC) are **BLOOD PRODUCTS** and require **written consent for transfusion** as per the [Transfusion Consent Policy](#), and must be prescribed on the MR826 Blood Product Transfusion Form.
3-point patient ID check is essential at every step in the transfusion process.

Please refer to the Transfusion Policy Manual, [Administration of Clotting Factors](#), for PCC product information and administration guidelines.

Withholding warfarin doses

Withholding warfarin will allow an elevated INR to slowly drift back into the target range. It may be the only intervention required. Note that the INR reflects warfarin dosing over several days. Therefore, it can be appropriate to restart warfarin at a lower dose when the INR is down-trending, but before it reaches the therapeutic range.

Vitamin K (Phytomenadione) administration

Vitamin K reverses the effects of warfarin. It can be given by an oral or intravenous (IV) route. Patients who require semi-urgent surgery (e.g. fractured neck of femur), should be given IV vitamin K as early as possible to minimise the need for reversal agents, all of which have associated risks. Oral vitamin K onset of action is ~ 4-6 hours compared to the faster onset of ~1-3 hours with the IV. The half-life of vitamin K is shorter than that of warfarin, so the INR may rise again after the administered vitamin K wears off. Daily INR monitoring is recommended.

The vitamin K dose can range from 0.5-10 mg. Dosing decisions depend on clinical factors such as:

1. Presence, site, and severity of bleeding.
2. Reason for INR elevation (e.g. interacting medication where, if left untreated, the INR may be expected to rise further).
3. Indication for warfarin and urgency of re-initiation (e.g. in patients with metallic mitral valve, where warfarin is the preferred anticoagulant and the risk of is relatively high).

Human Prothrombin Complex Concentrate (PCC)

Australia transitioned from 3-factor PCC (3FPCC; Prothrombinex VF®) to 4-factor PCC (4FPCC; Beriplex®) in the third quarter of 2024.

- 4FPCC contains factors II, VII, IX and X as well as protein C and protein S.
- Beriplex® is dosed according to factor IX concentration. Each vial is assigned a concentration of 500IU factor IX.
- Dose may be rounded to the nearest product vial size.

For **immediate** warfarin reversal, human prothrombin complex concentrates (PCC) are preferred over fresh frozen plasma (FFP).

- FFP is not needed in combination with PCC.
- FFP may be used when PCC is not available, is contraindicated (e.g. HIT, DIC) or in situations where there is concern regarding deficiency of additional clotting factors (e.g. concurrent severe liver disease, massive haemorrhage).
- Vitamin K is essential for sustaining the reversal achieved by PCC in patients who have been on warfarin.
- Correction of INR is commonly reached approx. 30 mins post PCC administration
- Repeated dosing of PCC for patients requiring urgent warfarin reversal is NOT recommended

 **Note that PCC contains small amounts of heparin.** Carefully consider its use in patients with a history of heparin induced thrombocytopenia (HIT).

1.0 Management of patients on warfarin therapy with bleeding

Multiple interventions are required for warfarin reversal in the bleeding patient. These include withholding warfarin, providing rapid and local haemostatic measures, administration of vitamin K, and, where required, the administration of PCC. Table 1 provides the current guidelines for management of patient on warfarin therapy with active bleeding.

Clinical setting	Recommendations
INR ≥ 1.5 with life-threatening or critical organ bleeding	Omit warfarin therapy and administer: • vitamin K 5.0–10.0 mg IV; and • 4FPCC 50 units/kg*
INR ≥ 2.0 with clinically significant bleeding (not life-threatening)	Omit warfarin therapy and administer: • vitamin K 5.0–10.0 mg IV; and • 4FPCC 25–50 units/kg based on INR and patient level factors (e.g. type of bleeding, need and ability to undergo intervention)
Any INR with minor bleeding	• Omit warfarin • Repeat INR and adjust warfarin dose to maintain INR in the target therapeutic range • If bleeding risk is high or INR > 4.5, consider vitamin K (1.0–2.0 mg orally, or 0.5–1.0 mg IV)
4FPCC = 4- factor prothrombin complex concentrate; * Consider administering a 4FPCC dose < 50 units/kg when INR 1.5–1.9.	

Table 1. Management of patients on warfarin therapy with bleeding¹

2.0 Management of patients requiring warfarin reversal for surgery

 Management of warfarin therapy peri-operatively should consider the competing risks of thromboembolism and bleeding, as well as determining the time of anticoagulation interruption and whether bridging anticoagulation is required.

Key points include:

- Surgery can be conducted with minimal increased risk of bleeding if INR ≤ 1.5 .
- For minor procedures where the risk of bleeding is low and sites compressible, warfarin may not need to be interrupted.
- Due to the associated prothrombotic risk, PCC should **not** be used to enable elective or non-urgent surgery.

Additional guidance includes:

- Patients having elective surgery should be managed such that PCC is **not** required.
- For surgery that **can be delayed 6-12 hours**, the INR should be corrected by giving IV vitamin K and rechecked before proceeding.
- For surgery that cannot be delayed, the INR should be corrected using IV Vitamin K and PCC. See Table 2 for dose suggestions. For confirmation of corrected INR, recheck INR 30 minutes post completion of administered dose.

For patients weighing more than 100kg, the dose of PCC should be calculated based on a capped weight of 100kg and a maximum dose of 50 units/kg (i.e. maximum dose of 5000 IU).

Target INR	Starting INR			
	1.5–2.5	2.6–3.5	3.6–10.0	> 10.0
0.9–1.3	30 units/kg	35 units/kg	50 units/kg	50 units/kg
1.4–2.0	15 units/kg	25 units/kg	30 units/kg	40 units/kg

Table 2. Suggested dose of 4FPCC to reverse warfarin anticoagulation in a non-bleeding patient (e.g. before surgery)¹

⚠ Outside the peri-operative setting, 4FPCC use should only be considered if the INR is >10 and there is a high risk of bleeding.

Issue of PCC

SCGH: contact the transfusion medicine unit (6383 4018) for release of PCC (Beriplex®). This will be released without haematologist authorisation for warfarin reversal.

OPH: contact Pathwest OPH (6457 8089). For critical bleed events/after hours, PCC is available in the theatre blood fridge.

It is the prescribing doctor's responsibility to ensure appropriate prescribing and use, and to seek haematology advice for complex cases or uses that are outside this guideline.

Compliance, monitoring and evaluation

Use of PCC will be audited and reviewed through the SCGOPHCG Blood Management Committee.

Related Documents

- [Records Management Act 2000](#)

Authorising Policy and Standard/s:

- [Clinical Governance, Safety and Quality Policy Framework](#)
- NSQHS Standard 7 Blood Management

Supporting documents

- Anticoagulation Medication Chart SCGH (MR 401/805.2), OPH (MR 172)
- Guidelines for anticoagulation using warfarin (Department of Health, 2019)
- Blood Product Transfusion Form (MR826/ MR(OPH)179)
- [SCGOPHCG Administration of Clotting Factors](#)
- [Beriplex® P/N Product Information](#)
- [Beriplex® AU Product Information](#)
- [Prothrombinex® -VF Product Information](#)

References

1. Robinson, D.,McFadyen, J., Merriman, E., et al. Updated recommendations for warfarin reversal in the setting of four-factor prothrombin complex concentrate. Medical Journal of Australia. 2024 Nov 25. Available from: <https://onlinelibrary.wiley.com/doi/10.5694/mja2.52538>
2. Aldhaeefi, M., Badreldin, H.A., Alsuwayyid, F., et al. Practical guide for anticoagulant and antiplatelet reversal in clinical practice. Pharmacy. 2023 Feb 11; 11(1). Available from: <https://www.mdpi.com/2226-4787/11/1/34>

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